

HubSpot Sync - Technical Submission Document

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1. Project Links

Item	URL
GitHub Repository	https://github.com/Yashvar2004/assignment
Frontend (Vercel)	https://hubspot-sync-frontend.vercel.app
Backend (Vercel)	https://hubspot-sync-backend.vercel.app

2. How to Use the Deployed Application

Step 1: Get Your HubSpot PAT Token

1. Go to <https://app.hubspot.com>
2. Click **Settings** (gear icon, top right corner)
3. In the left sidebar, go to **Integrations** → **Private Apps**
4. If you don't see "Private Apps", search for "**Legacy Apps**" in the settings search bar
5. Click "**Create a private app**"
6. Give it a name: `My HubSpot Sync App`
7. Go to the **Scopes** tab and add these scopes:
 - `crm.objects.contacts.read`
 - `crm.objects.contacts.write`
 - `crm.objects.companies.read`
 - `crm.objects.deals.read`
 - `crm.objects.deals.write`
8. Click "**Create app**"
9. Copy the **Access Token** (starts with `pat-na2-...`)

Step 2: Sign Up on the Deployed App

1. Open <https://hubspot-sync-frontend.vercel.app>
2. Click "**Don't have an account? Sign up**"
3. Enter your **email** and **password** (minimum 6 characters)
4. Click "**Create Account**"
5. You will see the "**Connect HubSpot**" page
6. Paste your **PAT token** (from Step 1) in the input field
7. Click "**Connect HubSpot**"
8. Wait **10-15 seconds** — your contacts will sync automatically

Step 3: Use the App

- **View contacts:** Your HubSpot contacts appear automatically
- **Add notes:** Click on any contact, type a note, and click "Add Note"
- **Notes sync to HubSpot:** Notes you add in the app appear in HubSpot automatically

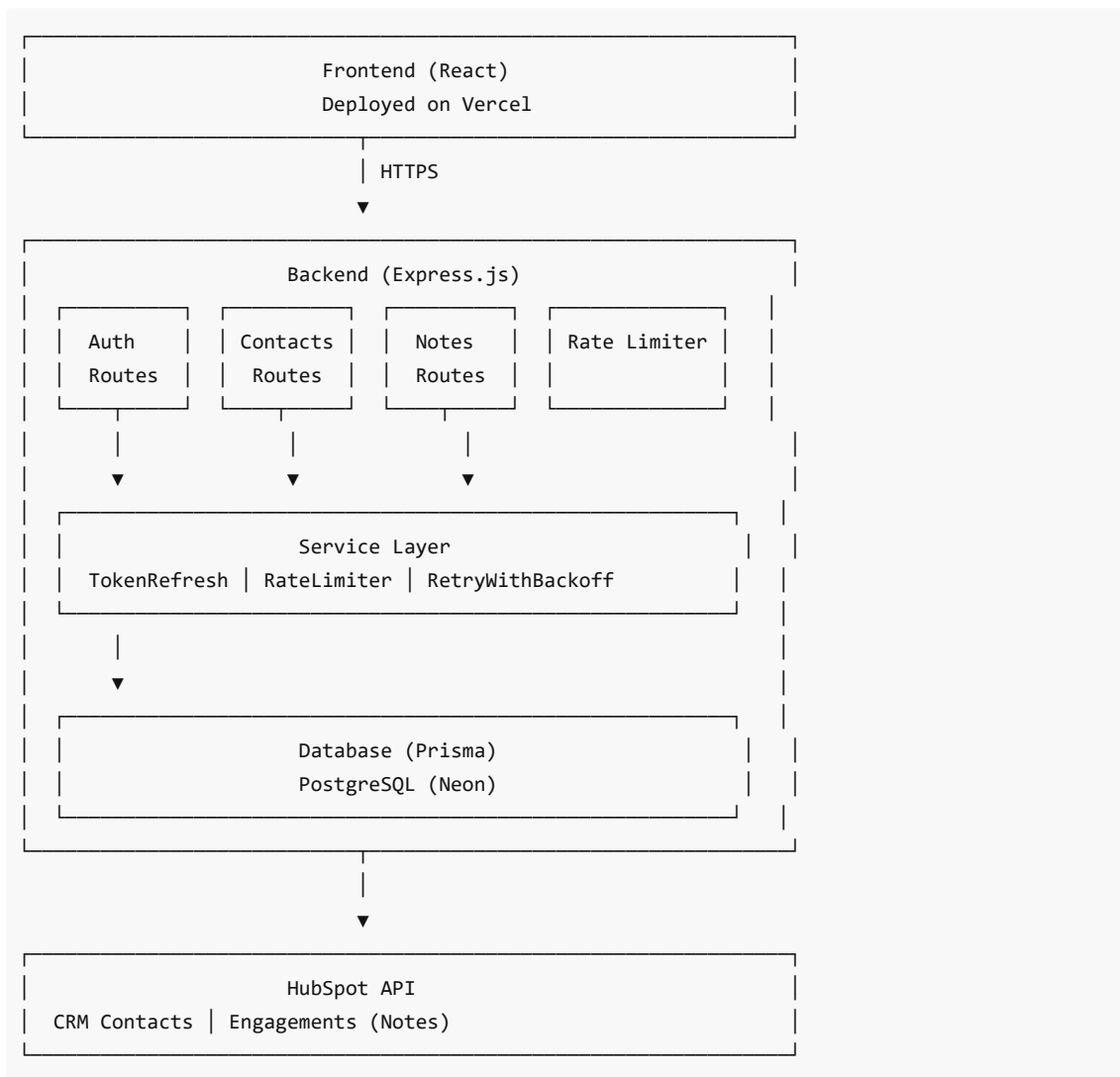
- **Auto-sync:** New contacts from HubSpot sync every 30 seconds
- **Refresh:** Click the "Refresh" button for instant sync

3. Project Overview

This application provides a production-ready integration with HubSpot's CRM platform:

- **User Authentication** — Register and login with email/password
- **HubSpot Integration** — Connect your HubSpot account using a PAT token
- **Automatic Contact Synchronization** — Contacts sync automatically after connection
- **Bidirectional Note Sync** — Notes created in the app sync to HubSpot
- **Auto-refresh** — Contacts sync every 30 seconds automatically
- **Multi-user Support** — Each user has their own contacts and data

4. Architecture



5. Technology Stack

Backend

Technology	Version	Purpose
Node.js	22.x	Runtime environment
Express.js	4.21.x	Web framework
Prisma	6.9.x	ORM for database
PostgreSQL	15.x	Database (Neon)
express-rate-limit	7.5.x	API rate limiting
jsonwebtoken	9.0.x	JWT authentication
axios	1.7.x	HTTP client with retry

Frontend

Technology	Version	Purpose
React	18.x	UI framework
TypeScript	5.7.x	Type safety
Vite	6.x	Build tool
Tailwind CSS	4.x	Styling
React Router	6.x	Client-side routing
Axios	1.7.x	HTTP client

6. Database Schema

User Table

```
CREATE TABLE User (  
  id          TEXT PRIMARY KEY,  
  email       TEXT UNIQUE,  
  password    TEXT,  
  name        TEXT,  
  hubspotPortalId TEXT,  
  portalName  TEXT,  
  accessToken TEXT,  
  refreshToken TEXT,  
  tokenExpiresAt TIMESTAMP,  
  scopes      TEXT,  
  createdAt   TIMESTAMP DEFAULT NOW(),  
  updatedAt   TIMESTAMP  
);
```

Contact Table

```
CREATE TABLE Contact (  
  id          TEXT PRIMARY KEY,  
  hubspotId   TEXT,  
  userId      TEXT REFERENCES User(id),  
  email       TEXT,  
  firstName   TEXT,  
  lastName    TEXT,  
  phone       TEXT,  
  company     TEXT,  
  jobTitle    TEXT,  
  lifecycleStage TEXT,  
  city        TEXT,  
  country     TEXT,  
  hsCreatedAt  TIMESTAMP,  
  hsUpdatedAt  TIMESTAMP,  
  lastSyncedAt  TIMESTAMP DEFAULT NOW(),  
  createdAt    TIMESTAMP DEFAULT NOW(),  
  updatedAt    TIMESTAMP,  
  UNIQUE(userId, hubspotId)  
);
```

Note Table

```
CREATE TABLE Note (  
  id          TEXT PRIMARY KEY,  
  hubspotEngagementId TEXT UNIQUE,  
  contactId   TEXT REFERENCES Contact(id),  
  body        TEXT,  
  syncedToHubspot BOOLEAN DEFAULT FALSE,  
  syncAttempts INT DEFAULT 0,  
  lastSyncError TEXT,  
  lastSyncAttempt  TIMESTAMP,  
  createdAt    TIMESTAMP DEFAULT NOW(),  
  updatedAt    TIMESTAMP  
);
```

SyncJob Table

```
CREATE TABLE SyncJob (  
  id          TEXT PRIMARY KEY,  
  userId      TEXT REFERENCES User(id),  
  type        TEXT,  
  status      TEXT DEFAULT 'pending',  
  totalItems  INT,  
  processed   INT DEFAULT 0,  
  failed      INT DEFAULT 0,  
  cursor      TEXT,
```

```
    startedAt    TIMESTAMP,  
    completedAt  TIMESTAMP,  
    error        TEXT,  
    createdAt    TIMESTAMP DEFAULT NOW(),  
    updatedAt    TIMESTAMP  
);
```

7. API Documentation

Authentication

Method	Endpoint	Description
POST	/api/auth/register	Register new user
POST	/api/auth/login	Login
GET	/api/auth/me	Get current user
POST	/api/auth/connect-hubspot	Connect HubSpot with PAT token
GET	/api/auth/status	Check connection status
POST	/api/auth/disconnect	Disconnect HubSpot

Contacts

Method	Endpoint	Description
GET	/api/contacts	List contacts (paginated, searchable)
GET	/api/contacts/:id	Get contact details
POST	/api/contacts/sync	Trigger contact sync
GET	/api/contacts/sync/jobs	Get sync jobs
GET	/api/contacts/sync/jobs/:jobId	Get sync job status

Notes

Method	Endpoint	Description
POST	/api/contacts/:contactId/notes	Create note
GET	/api/contacts/:contactId/notes	Get notes
DELETE	/api/notes/:noteId	Delete note
GET	/api/notes/sync-status	Note sync status
POST	/api/notes/retry-sync	Retry failed note syncs

Health

Method	Endpoint	Description
GET	/health	Health check

8. Scalability Design

Feature	Implementation
Rate Limiting	express-rate-limit (100 req/15min) + HubSpot token bucket (100 req/10s)
Retry	Exponential backoff with jitter (5 attempts)
Idempotent Operations	Upsert with compound unique constraint
Cursor-based Pagination	Resumable syncs with cursor tracking
Batch Processing	Promise.all for parallel upserts
Token Refresh	Auto-refresh 5 minutes before expiry

Production Scaling Recommendations

1. **BullMQ + Redis:** Extract background jobs to dedicated workers
2. **Database Read Replicas:** Separate read and write databases
3. **Caching Layer:** Redis cache for frequently accessed data
4. **Multiple Worker Instances:** Deploy workers across servers

9. Limitations

1. **PAT Token Authentication:** Uses PAT token instead of full OAuth flow (HubSpot disabled public app creation)
2. **In-memory Rate Limiting:** Rate limits reset on serverless cold start
3. **No Queue System:** Background jobs run in-process
4. **Single HubSpot Account per User:** Each user connects one HubSpot account

10. Future Improvements

1. **BullMQ + Redis:** Extract background jobs to dedicated workers
2. **Full OAuth Flow:** Implement when HubSpot re-enables public app creation
3. **Webhook Support:** Real-time updates from HubSpot
4. **Incremental Sync:** Only sync changed contacts
5. **Docker Support:** Containerized deployment
6. **Unit Tests:** Comprehensive test coverage

11. Contact Information

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